DSAIT4100 NLP for Society
Project Proposal

Group 3: Atharva Dagaonkar, Ahmed Driouech, Alexandru-Valentin Florea, Riya Gupta

Domain: Implicit vs Explicit Hate Speech Detection in a Multilingual Setting

Hate speech detection is a sensitive topic; false negatives leave targeted communities unprotected, while false positives suppress legitimate speech. Research in this domain distinguishes explicit hate, which uses slurs, dehumanising, or negative attributes, from implicit hate, which conveys hostility through stereotyping or coded language without lexical cues a keyword system would catch [1]. Implicit hate is especially hard to predict. El Sherief et al. [1] report substantial drops in performance on implicit instances, and recent work shows that even instruction tuned Large Language Models (LLMs) often over predict hate on neutral content with identity terms while under predicting hostile statements [6].

However, most of these studies only base their research on English datasets. Recent work has evaluated LLMs on functional benchmarks such as Multilingual HateCheck (MHC) [3] across multiple languages [5, 2], but these studies do not make a distinction between implicit and explicit hate speech in their experiments. The MHC dataset does allow for distinction between implicit and explicit hate speech instances, but no study in this domain seems to address this gap. Therefore, this project addresses that gap with an implicit vs explicit decomposition of LLM behaviour across all MHC languages, with English as baseline.

Dataset

We use the **Multilingual HateCheck (MHC)** dataset [3], a diagnostic suite of functional tests for hate speech detection models covering 10 languages (Arabic, Dutch, French, German, Hindi, Italian, Mandarin, Polish, Portuguese, Spanish) in addition to the original English HateCheck [4]. MHC provides fine-grained *functionality* labels that allow us to operationalise the implicit–explicit distinction. We group the following functionalities into two categories:

- **Explicit hate:** `derog_neg_attrib_h` (derogatory negative attribute), `derog_dehum_h` (dehumanization), `slur_h` (slur usage), `profanity_h` (hateful profanity).
- **Implicit hate:** `derog_impl_h` (implicit derogation) [1].
- **Non-hateful control:** `profanity_nh` (profanity in a non-hateful context).

This grouping allows controlled comparison of model performance on implicit vs. explicit categories across all 11 languages.

Research Questions

RQ1: How does hate speech detection performance differ between implicit and explicit hate across languages?

SQ1.1: What is the performance gap (F1, precision, recall) between implicit (`derog_impl_h`) and explicit (`slur_h`, `derog_dehum_h`, `derog_neg_attrib_h`, `profanity_h`) hate for a fine-tuned multilingual encoder (XLM-RoBERTa)?

SQ1.2: How does this gap vary across the 11 languages in MHC [3]? Are certain languages disproportionately affected?

RQ2: Can generating natural language explanations for hate labels improve classification accuracy, particularly for implicit hate?

SQ2.1: Does a pipeline that jointly predicts a hate label and generates an explanation of the hateful intent (using a prompted LLM) outperform a classification-only baseline on implicit hate instances?

SQ2.2: Does explanation generation provide greater gains for implicit hate (`derog_impl_h`) than for explicit categories, where lexical signals are already strong?

SQ2.3: How does the quality of generated explanations (evaluated via human judgment or a reference-free metric) correlate with classification correctness?

RQ3: How do encoder-based and LLM-based approaches compare in their robustness to the implicit–explicit distinction across languages?

SQ3.1: How does zero-shot or few-shot prompting of an LLM (e.g., a Llama or Mistral variant) compare to fine-tuned XLM-RoBERTa on implicit vs. explicit hate across MHC languages?

SQ3.2: For which functionality–language combinations does the LLM-based approach show the largest advantage or disadvantage relative to the encoder baseline?

SQ3.3: Does providing the LLM with a structured explanation prompt (e.g., "Classify this text and explain why it is or is not hateful") reduce the implicit–explicit performance gap more effectively than a direct classification prompt?

Resources

- **Dataset:** Multilingual HateCheck (`https://huggingface.co/datasets/mteb/multi-hatecheck`)
- **Training data:** HASOC shared task datasets (English, Hindi, German) or a combined multilingual hate speech corpus for fine-tuning.
- **Models:** XLM-RoBERTa (base or large), an open-weight LLM (e.g., Mistral-7B or Llama-3-8B).

Plan of Action

Methods

RQ1: Implicit vs. Explicit Performance Gap Across Languages

XLM-RoBERTa (base or large) will be fine-tuned on a combined multilingual hate speech corpus (e.g., HASOC shared task data) for binary hate speech classification, then evaluated directly on MHC as a diagnostic benchmark. Per-functionality F1, precision, and recall will be computed for the implicit category (`derog_impl_h`) and each explicit category across all 11 languages, producing a functionality $\times$ language results matrix from which the implicit–explicit gap and any language-level disparities can be quantified.

RQ2: Explanation-Augmented Classification

An open-weight LLM (e.g., Mistral-7B or Llama-3-8B) will be prompted to jointly assign a hate label and generate a natural language rationale. Classification performance is compared against the XLM-RoBERTa baseline with separate breakdowns for implicit and explicit categories. Explanation quality will be assessed via a reference-free metric (e.g., BERTScore or G-Eval), and its correlation with classification correctness examined.

RQ3: Encoder vs. LLM Robustness Across Languages

Zero-shot and few-shot prompting of the LLM will be compared against fine-tuned XLM-RoBERTa across the full functionality $\times$ language matrix. Two prompt strategies will be contrasted: a direct classification prompt and a structured explanation prompt. Analysis will focus on functionality–language combinations where the two approaches diverge most, with particular attention to implicit hate and lower-resource MHC languages.

Phases

The work will be structured in four phases.

1. **Data preparation, baseline, and literature review.** The MHC dataset is downloaded and preprocessed, the explicit/implicit category groupings are verified, and the fine-tuning corpus is assembled. XLM-RoBERTa is fine-tuned and an initial evaluation on MHC is run to confirm the pipeline is functioning correctly. Concurrently, a literature survey is conducted covering hate speech detection, the implicit–explicit distinction, multilingual NLP, and explanation-guided classification.

2. **RQ1 and RQ3 encoder experiments.** Full per-functionality, per-language evaluation of XLM-RoBERTa is completed and the results matrix is populated. Zero-shot and few-shot LLM prompting experiments are run in parallel, and the cross-model comparison is assembled.

3. **RQ2 explanation experiments.** The explanation-augmented pipeline is implemented and evaluated. Explanation quality assessment is carried out on a sampled subset. The correlation analysis between explanation quality and classification correctness is completed.

4. **Analysis and report.** A qualitative error analysis is conducted on failure cases, with a focus on implicit hate in linguistically diverse languages. The report is drafted and revised, and the replication package is cleaned, documented, and verified to run end-to-end.

Task division

1. Atharva Dagaonkar:
   - Data preparation & preprocessing: Verify and group MHC functionality labels into explicit hate, implicit hate, and non-hateful control categories
   - Responsible for preprocessing the Multilingual HateCheck (MHC) dataset and assembling the multilingual training corpus from HASOC and related hate speech datasets
   - Evaluation support: Assist with defining the evaluation metrics and collaborating with the evaluation team for this and to design functionality-level and language-level tests
2. Alexandru-Valentin Florea:
   - Baseline model development: Responsible for implementing, fine-tuning, and optimizing the XLM-RoBERTa baseline model for multilingual hate speech classification
   - Develop the training and inference pipeline, including dataset integration, tokenization, and reproducibility setup
   - Handle hyperparameter tuning, architecture configuration, and optimization of the encoder-based classification framework
3. Ahmed Driouech:
   - Responsible for implementing (instruction formatting, parameter tunning etc) and fine-tuning open-weight LLMs such as Mistral-7B or Llama 3 for multilingual hate speech classification
   - Prompt engineering: Design and refine zero-shot, few-shot, and explanation-based prompting strategies for implicit and explicit hate speech detection
4. Riya Gupta:
   - Evaluation & experimentation: Responsible for conducting encoder and LLM evaluation experiments in all MHC languages and functionalities
   - Compute and analyze F1, precision, and recall scores for implicit and explicit hate categories
   - Evaluate generated explanations using reference-free metrics (for eg. BERTScore or G-Eval) and analyze correlations with classification correctness
   - Results interpretations, visualizations
5. Shared:
   All group members will contribute to literature review, discussion of findings, presentation preparation, and revision of the final report.

References

[1] Mai ElSherief, Caleb Ziems, David Muchlinski, Vaishnavi Anber, Jordan Gregoire, Joleen Nham, and Diyi Yang. Latent hatred: A benchmark for understanding implicit hate speech. In *Proceedings of EMNLP 2021*, 2021.

[2] Faeze Ghorbanpour, Daryna Dementieva, and Alexander Fraser. Can prompting LLMs unlock hate speech detection across languages? A zero-shot and few-shot study. In *Proceedings of the 9th Workshop on Online Abuse and Harms (WOAH)*, Vienna, Austria, 2025. Association for Computational Linguistics.

[3] Paul Röttger, Haitham Seelawi, Debora Nozza, Zeerak Talat, and Bertie Vidgen. Multilingual HateCheck: Functional tests for multilingual hate speech detection models. In *Proceedings of the 6th Workshop on Online Abuse and Harms (WOAH)*, 2022.

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[5] Manuel Tonneau, Diyi Liu, Niyati Malhotra, Scott A. Hale, Samuel Fraiberger, Victor Orozco-Olvera, and Paul Röttger. HateDay: Insights from a global hate speech dataset representative of a day on twitter. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2297–2321, Vienna, Austria, 2025. Association for Computational Linguistics.

[6] Min Zhang, Jianfeng He, Taoran Ji, and Chang-Tien Lu. Don't go to extremes: Revealing the excessive sensitivity and calibration limitations of LLMs in implicit hate speech detection. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12073–12086, Bangkok, Thailand, 2024. Association for Computational Linguistics.